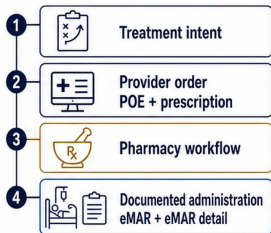


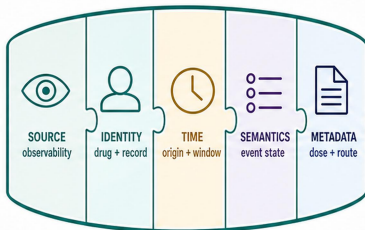
Medication exposure is a resolved joint provenance operator—not a table

1 CLINICAL DATA GENERATION



Distinct provenance layers—
not interchangeable records

2 JOINT PROVENANCE OPERATOR

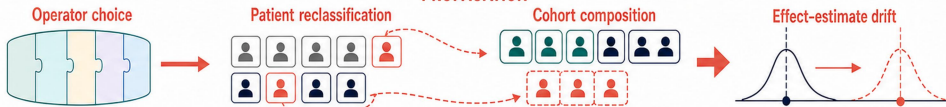


Documented administration is not biological exposure

3 ANALYTIC EXPOSURE



PROPAGATION



Coral indicates discordant or reclassified patients



Raw clinical records (provenance sources)



Pharmacy / time layer



Resolved exposure output



Not resolved / unresolved